

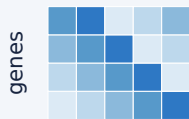
GoCo workflow

1 Predictor outputs

gene-GO scores

S_{ig}

sources behind each
candidate call



GO terms

Data split

ranking T: 10%
certification C: 60%
evaluation E: 30%

**uniform random
partition**

2 Candidate policies

**Nested paths using
predictor-native
structure**

GoCo-M

shared module evidence
partial admission
of genes within grid steps

GoCo-N

gene-specific evidence
from neighbourhoods
individual-call ordering

Paths fixed using T
before certification

3 Certification on C

Test candidate policies
in path order

$\pi_1, \pi_2, \dots, \pi_K$

**Fixed-sequence
Learn-Then-Test**

Stop at first $p_k > \delta$

Proposition 1

$\Pr \{R(\hat{\pi}) > \alpha\} \leq \delta$
asymptotically

under stated assumptions

4 Common evaluation on E

Held-out gene-level
TruePath FDP

Reference-supported
released calls



yield and risk diagnostics

**Same frozen reference
for every method**